

Coding with Generative AI in Data Science

Data Science Workshop · August 2026



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Overview

Topics Covered Today

Responsible Use

- Data Privacy & Ownership
- Transparency When Using AI
- Values-Based Decision Making

Background

- What is Generative AI
- Developments in AI
- Types of AI Platforms/Models
- AI Strengths & Weaknesses
- Prompting Frameworks
- Current Cautions

Use in Data Science

- Log into CatChat
- Research Workflow
- Research Questions

Use Cases

- Generate Data
- Data Cleaning
- Data Summarization
- Visualization
- Analysis Methods
- Analysis Execution
- Interpreting Outputs
- Translating Code
- Documentation
- Debugging Errors

More Tools

- Additional Uses Cases
- Additional AI Tools

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Responsible Use

Data Privacy and Ownership

Human Subjects

- Does the data you are entering into Gen AI have identifiers like names, birth dates, addresses, etc.? Or is it sensitive in other ways?

Intellectual Property

- Does the data you're entering into Gen AI belong to anyone?
- Did someone write the words or create the images?

Terms of Service

- What are the privacy and data use settings?
 - Will the company sell the data?
 - Use it to train new models?
 - Use it as sample data for other users?

Non-Human Sensitive Data

- Geotagged archaeological site data
- Endangered species data
- Copyrighted data

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Responsible Use Data Privacy and Ownership

Levels of Privacy

Tool	Use with Public Data	Use with Restricted Data (PII, FERPA, ect.)	Use with Confidential Data (SSN, ect.)
Public tools (personal copilot, chatGPT, ect.)	Allowed if licensed but not supported	No	No
Copilot (free or paid) when signed in with MSU account	Yes	Yes	No
CatChat	Yes	Yes	No
Tempest HPC	Yes	Yes	No

<https://www.montana.edu/uit/rci/catchat/catchat.html>

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Responsible Use Transparency When Using AI

If you do decide to use AI, proper citation practices can help facilitate responsible use of AI.

Include

- Name the use or function AI provided to your work
- Vet sources generated by AI
- Name the tool
- Where it is used in your work
- Date the content was generated
- Document in the code

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Responsible Use Values-Based Decision Making

University Values

- MSU has guidelines for academic integrity that are outlined in our Code of Student Conduct. Although AI is a new technology, its use should still align with expectations of academic integrity.
- The university values student learning, and it expects students to “be responsible for the honest completion and representation of their work, the appropriate citation of sources, and the respect and recognition of others' academic endeavors” (Code of Student Conduct, Section 200.00).

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Responsible Use Values-Based Decision Making

Professor Values

- Talk with your professors about their specific expectations regarding using AI for homework and studying.
- This resource from the Center for Faculty Excellence provides information about how the university and your professors may be thinking about AI.
<https://www.montana.edu/facultyexcellence/teaching-advising/courseclassdesign/genai/index.html>

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Responsible Use Values-Based Decision Making

Your Values

- Which of your personal values relate to AI?
 - Critical Thinking
 - Convenience
 - Climate Considerations
 - Information Trustworthiness
 - Innovation

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Responsible Use Values-Based Decision Making

Your Values

- Which of your personal values relate to AI?
- Does using AI help you build on or process your own ideas?
- Do you value learning and new challenges?
- Are you concerned about receiving inaccurate or biased information from an AI source that might affect your success in the classroom?
- Do you value efficiency, using AI to reduce the amount of time you spend on assignments?

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Responsible Use More Information

Library guide on AI

<https://guides.lib.montana.edu/ai>

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Background

What Is Generative AI

- A type of artificial intelligence that creates new outputs based on patterns learned from existing data
- Trained on massive datasets and can be prompted to generate novel outputs that resemble the patterns in data they were trained on
- Users provide a "prompt" (a text description, an image, etc.), and the AI generates a response (can produce text, code, images, audio, etc.)
- LLMs (Large Language Models): trained on and focus on language/writing
- Like a very advanced "autocomplete"
- Like a junior assistant, not an independent researcher

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Background

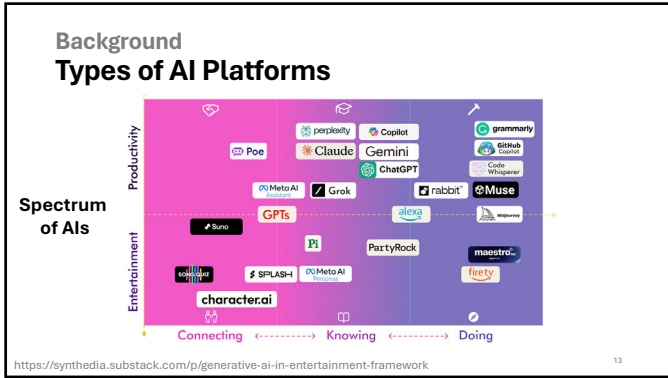
Developments in AI

Biggest Shifts

- Moving from chatbots ("chat with a model to get an answer") toward agents ("give AI a task and let it work through the task")
- Increasing use throughout entire research workflow
- From "can AI do this" to "how do I verify outputs"
- AI can now execute and iterate on code
- AI literacy becoming part of research literacy

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Background

Types of AI Models

Use Case	Requirements	Recommended Models
Advanced reasoning and multi-domain research	Highest accuracy, long context reasoning, strong factual recall, and cross-domain synthesis (science, engineering, policy, etc.)	gpt-oss-120b
Enterprise copilots / knowledge assistants	Strong reasoning, safe output, high reliability, API latency tolerance	gpt-oss-120b , gemma3-27b
General-purpose chatbots and customer service	Balanced cost-performance, multilingual fluency, context awareness	gemma3-27b , mistral-small-24b
Code generation and debugging (multi-language)	Syntax awareness, structured reasoning, efficiency on moderate hardware	gemma3-27b , mistral-small-24b
Lightweight internal automation (summarization, tagging, extraction)	Low latency, scalable inference, moderate reasoning depth	mistral-small-24b , gemma3-9b
Fine-tuning / domain adaptation experiments	Model that trains efficiently, flexible licensing	mistral-small-24b , gemma3-9b
Creative writing, marketing copy, dialogue	Natural tone generation, stylistic diversity, moderate context memory	gemma3-27b , mistral-small-24b
Technical document summarization	Structured understanding, moderate token context, reliability	gemma3-27b , gpt-oss-120b
Research prototyping / evaluation of LLM benchmarks	Open weights, transparency, large context	gpt-oss-120b , mistral-small-24b

<https://www.montana.edu/uit/rci/catchat/catchat.html>

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Background

AI Strengths & Weaknesses

Strengths

- Writing and translating code
- Generating, transforming, and summarizing information
- Explaining concepts and helping you learn
- Brainstorming ideas and possible approaches
- Writing, translating, and editing
- Generating and modifying code
- Finding patterns and suggesting possibilities
- Handling well-defined, repeatable tasks
- Iterating quickly based on feedback

Weaknesses

- Knowing when it is wrong
- Understanding context it hasn't been given
- Making decisions requiring expertise or judgment
- Determining what is true or trustworthy
- Reliably interpreting ambiguous situations
- Distinguishing subtle technical interpretations
- Knowing what information is important or missing

Yes: Well-defined, bounded, verifiable tasks

Caution: Tasks involving interpretation or methodological choices

No: Tasks where correctness depends heavily on scientific context, judgment, or verification

- The easier an AI-generated result is to verify, the safer it is to delegate
- Different concerns/uses than when used for academics or writing

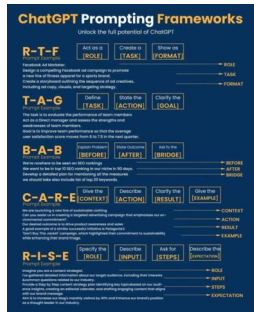
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Background Prompting Frameworks

- Write good prompts >> give AI good context and specifications
- Quality of what you provide still strongly affects quality of result
- Context more important than wording
- More context not automatically better

Good AI Interaction

- **Goal:** What are you trying to accomplish?
- **Context:** What is the study/data/problem?
- **Inputs:** Give relevant data, code, files, or output
- **Constraints:** What should it preserve or avoid?
- **Expected output:** What do you want back?
- **Examples:** Show it what you mean when useful
- **Feedback:** Tell it what was wrong and let it revise



<https://inkd.in/p/gYwIk2dc>

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Background Current Cautions

General

- Privacy
- Bias
- Overconfidence
- Reproducibility
- Hallucinations
- Over-reliance / Skill atrophy

Technical

- Prompt injection / Security
- Stopsquatting
- Model collapse
- Data poisoning
- Code bloat / Technical debt



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Use In Data Science Log into CatChat

<https://catchat.montana.edu/>

Continue with Microsoft

- Continue with Microsoft
- Login with university credentials

If don't have university credentials:
Claude, ChatGPT, Copilot, Gemini, etc.



<https://www.montana.edu/ait/rci/catchat/catchat.html>

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Use In Data Science

Research Workflow

Before Analysis

- Brainstorming research questions
- Finding/searching literature
- Understanding methods
- Planning analyses
- Thinking through study designs

During Analysis

- Synthesizing & documenting code
- Data cleaning
- Exploratory analysis
- Visualization
- Statistical explanations
- Interpreting output
- Debugging

After Analysis

- Writing
- Editing
- Summarizing
- Creating tables
- Preparing presentations
- Generating documentation

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Use In Data Science

Research Questions

Biologist or ecologist, health and abundance of fish in Blackfoot River

Population Health

- Does the health (length and weight) of different species of trout vary by river section, and does this correlate with the available food resources in those sections?
- Has the average size (length/weight) of the fish population increased or decreased over the study period?
- Do different trout species exhibit different length-to-weight growth curves in the same river environment?

Habitat Preference

- Is there a difference in the abundance of different trout species between the different river sections (indicating different temperature or flow preferences)?
- Which river section produces the "healthiest" fish (highest weight-to-length ratio), and what environmental factors in that section might be driving this?
- Do marked fish appear more frequently in certain sections than others?

Population Demographics

- Is there an increase or decline in the total number of fish captured per trip over the 17-year period?
- Is the population dominated by small, young fish or large, old fish? Has this age structure shifted over time (e.g., a lack of young fish suggesting a recruitment failure)?
- Is the ratio of different trout species shifting over time, suggesting that one species is out-competing the other?

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Use Cases

Generate Data

- I have data on fish from different sections of the Blackfoot River with variables trip, mark, length, weight, year, section, species. Can you generate data in R for this?

```

15.18 s | 82.68 T/s | 1255 Tokens | $0.000088
To generate a synthetic dataset that mimics the structure and distribution of your @blackfootfish dataframe, we can use the rnorm(), rsample(), and sample() functions in R.
Based on your summary, I have estimated the distributions (e.g., using the Mean and 1st/3rd Quartiles to approximate standard deviations for length and weight).
Here is the R code to generate a similar dataset:

R                                                                    Collapse Save Copy
1 # Set seed for reproducibility
2 set.seed(123)
3
4 # Define the number of observations (based on your summary totals)
5 # Summing species: 3171 + 553 + 12341 + 2287 = 18,352
6 n <- 18352
7
8 # 1. Trip: Binomial/Discrete (Range 1-2)

```

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Use Cases Data Cleaning

- Can you suggest data cleaning and R code for this data?

2. R Code for Data Cleaning

```
#
1 # Load necessary libraries
2 library(tidyverse)
3
4 # Assuming your dataframe is called df
5 # If using the simulated data from before
6
7 # 1. Initial Inspection
8 glimpse(blackfootfish)
9 summary(blackfootfish)
10
11 # 2. Data Cleaning Pipeline
12 blackfootfish_clean <- blackfootfish
13 # Convert categorical variables to
14 factors
15 section = as.factor(section),
16 species = as.factor(species),
17 trip = as.factor(trip)
18 } %>%
19
```

gemma4
10.07 x 141.83 Tm | 2222 Tokens | \$0.00140

Based on the summary statistics you provided, your dataset has a few common "real-world" data issues: **missing values (NAs)**, **extreme outliers** (specifically in weight), and **potential data entry errors** (e.g., a fish weighing 4.677g but with a mean of 246g).

Here is a comprehensive data cleaning workflow using the `tidyverse` package, which is the industry standard for data manipulation in R.

1. Recommended Cleaning Strategy

- Handle Missing Values:** Decide whether to remove rows with `NA` weights or impute them (fill them in) using the median weight of that specific species.
- Outlier Detection & Treatment:** The max weight (4.677) is nearly 20 times the mean. You should check if this is a biological possibility or a typo.
- Type Conversion:** Ensure `section` and `species` are treated as `factors` rather than `characters` for better analysis and plotting.
- Consistency Checks:** Ensure `length` and `weight` are logically consistent (e.g., a tiny fish shouldn't have a massive weight).

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Use Cases Data Summarization

- Can you summarize the data so I can look for differences?

3. Comparison by Year (Temporal)
Use this to see if the fish population

```
#
1 year_summary <- blackfootfish
2 group_by(year) %>%
3 summarise(
4   total_caught = n(),
5   avg_length = mean(length),
6   avg_weight = mean(weight),
7   # Calculate the proportion of fish marked
8   prop_marked = sum(mark == "marked") / n(),
9   # To see if there is a trend in marking
10  library(ggplot2)
11
```

2. Comparison by Section (Spatial)
Use this to see if certain parts of the

```
#
1 section_summary <- blackfootfish_clean
2 group_by(section) %>%
3 summarise(
4   count = n(),
5   avg_length = mean(length),
6   avg_weight = mean(weight),
7   # Calculate the proportion of fish marked
8   prop_marked = sum(mark == "marked") / n(),
9   # To see if there is a trend in marking
10  library(ggplot2)
11
```

1. Comparison by Species (Biological Differences)
Use this to see if one species is generally larger or more frequent than the other.

```
#
1 species_summary <- blackfootfish_clean
2 group_by(species) %>%
3 summarise(
4   count = n(),
5   avg_length = mean(length, na.rm = TRUE),
6   avg_weight = mean(weight, na.rm = TRUE),
7   avg_weight = mean(weight, na.rm = TRUE),
8   mark_rate = mean(mark == "marked", na.rm = TRUE) # Proportion of fish marked
9 )
10
11 print(species_summary)
```

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Use Cases Visualization

- What kind of plot can I make to explore research questions?
- Can you make a plot with height and weight by species over the years?

gemma4
10.01 x 291.43 Tm | 4881 Tokens | \$0.002089

Since you have a mix of continuous (length, weight), categorical (species, section), and temporal (year) data, you should use a variety of plots. I recommend using the `ggplot2` library in R.

Here are the best plot types mapped to the research questions we discussed:

1. For Health & Growth (Length vs. Weight)
The Plot: Scatter Plot with Regression Lines
This is the most important plot for a biologist. It shows the "growth curve." By adding a smooth line for each species, you can see if one species is "stouter" than the other.

```
R
1 ggplot(blackfootfish_clean, aes(x = length, y = weight, color = species)) +
2   geom_point(alpha = 0.3) +
3   geom_smooth(method = "lm", se = FALSE) +
```

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Use Cases

Analysis Methods

- What statistical tests should I use to test my research questions?

gemma4
17.32 s | 336.60 T/s | 5831 Tokens | \$0.000322

To choose the right statistical test, you must first look at the **type of data** you are comparing. In your dataset, you have **Continuous variables** (length, weight) and **Categorical variables** (species, section, trip).

Here are the statistical tests mapped to the research questions we developed, divided by the goal of the analysis.

1. Comparing Means (Differences between Groups)

Use these when you want to know if one group is "significantly" different from another (e.g., *Is Section A's fish larger than Section B's?*).

RESEARCH QUESTION	VARIABLE 1 (CATEGORICAL)	VARIABLE 2 (CONTINUOUS)	RECOMMENDED TEST	R FUNCTION
Two Species Comparison	species (2 levels)	length	Independent t-test	<code>t.test(length ~ species, data = df)</code>
Section Comparison	section (3+ levels)	weight	One-Way ANOVA	<code>oneway.test(weight ~ section, data = df)</code>

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Use Cases

Analysis Execution

- Can you do a linear regression of fish length over time, species, and section of river?

gemma4
13.71 s | 483.08 T/s | 6624 Tokens | \$0.000354

To do this, you will use a **Multiple Linear Regression** model. In R, the `lm()` function allows you to include several predictors (independent variables) to see how they simultaneously affect a response variable (fish length).

Since **Species** and **Section** are categorical variables, R will automatically treat them as "dummy variables," comparing each category to a baseline.

The R Code

```
#
# 1. Ensure categorical variables are factors
# This is critical for the regression model
BlackfootFish_Clean <- BlackfootFish
mutate(
  species = as.factor(species),
  section = as.factor(section),
  year = as.numeric(year)
)

# 2. Build the Multiple Linear Regression Model
# Formula: length ~ year + species + section
```

How to Interpret the Output

When you run `summary(fish_model)`, look at the **Coefficients** table. Here is how to read the results like a scientist:

- The Intercept**
The intercept is the predicted length when all other variables are at their baseline (e.g., Year 0, baseline species, baseline section). It usually isn't biologically meaningful on its own.
- The year coefficient (The Trend)**
Positive value: For every one-year increase, the average length of the fish increases by [X] mm.
P-value (< 0.05): If the p-value is small, there is a statistically significant trend in fish size over time.
- The species coefficients (The Biological Difference)**
You will see something like `speciesBull`.
Positive value: Bull sturgeon are, on average, longer than the baseline species (e.g., Blackfoot sturgeon).

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Use Cases

Interpreting Outputs

- What does a Multiple R-squared: 0.1693 mean for my model?
- What does an estimate of 1.8067 for year mean in context?

gemma4
18.28 s | 395.74 T/s | 7235 Tokens | \$0.000380

In the context of your linear regression model, an estimate (coefficient) of **1.8067** for `year` means:

For every one-year increase in time, the average length of the fish is predicted to increase by **1.8067** units (likely millimeters).

Here is the detailed breakdown of what that implies for your research:

- The Direction of the Trend**
Because the number is **positive**, there is a positive correlation between time and fish length. In plain English: **The fish in the Blackfoot River are getting longer over the study period.**
- The Magnitude of the Change**
To understand if 1.8067 is "a lot" or "a little," you have to look at the total timeframe. Your data spans from 1989 to 2006 (17 years).

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Use Cases

Translating Code

- A collaborator gave me some code they ran on the data. What does this do?
- Can you translate this code to R?

11.94 s | 670.46 Tps | 8004 Tokens | \$0.000473

The code your collaborator gave you is in SAS. It's a SAS program that creates a contingency table. Basically, your collaborator wants to know how many fish of each species were caught in each year. They want to see the raw counts, but they also want to see the percentages of the total fish caught in each year.

1. `proc contents data=fish_data;`

2. `proc freq (The Cross-Tabulation)`

```

sas
1 proc freq data=fish_data;
2   tables year*species / norow nocol nopercnt;
3 run;

```

• **What it does:** This creates a Contingency Table (a cross-tab). It counts how many fish of each species were caught in each year.

• **The Options (/ norow nocol nopercnt): These tell SAS to only show the raw counts. It suppresses the row percentages, column percentages, and total percentages to keep the table clean.**

• **R Equivalent:** `table(fish_data$year, fish_data$species)`

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Use Cases

Documentation

- Can you document the steps I have taken in this analysis?
- Can you document each line of code?

15.40 s | 578.86 Tps | 8915 Tokens | \$0.000473

Since you have moved from raw data simulation and cleaning to advanced statistical modeling and collaborating with other software users, it is important to have a formal "Analysis Pipeline."

Here is a professional documentation of the steps you have taken. You can copy this directly into a README file, a lab notebook, or the Methods section of a research paper.

Analysis Documentation: Blackfoot River Fish Population Study

1. Data Acquisition & Simulation

- Dataset: `BlackfootFish`
- Scope: A longitudinal study (1989-2006) of fish (Brown and Bull trout) across multiple sections of the Blackfoot River.
- Variables:
 - `trip`: Sampling event identifier.

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Use Cases

Debugging Errors

- I got this error, why and how do I fix it?

10.50 s | 909.62 Tps | 9555 Tokens | \$0.000496

This error is a classic R issue. It happens because R is trying to find a column named `Response Number`, but it cannot find it in your dataframe.

Looking at your previous code and the error message, there are two likely reasons why this is happening:

1. The "Space" Problem (Most Likely)

In R, if a column name has a space in it (like `Response Number`), you cannot refer to it normally. R thinks `Response` is the column name and `Number` is the variable name.

How to fix the code (Example)

Depending on which problem it is, here is how to rewrite your `mutate` line:

Option A: If the column has a space (use backticks):

```

R
1 BlackfootFish_Clean <- BlackfootFish %>%
2   mutate(across('`Response Number`', as.numeric)) # Use backticks!

```

Option B: If the column was renamed to dots:

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Use Cases

Learning/Explanations

- Can you explain `geom_text_repel()` and its parameters?

Sure — `geom_text_repel()` comes from the **ggrepel** package, and it works like `geom_text()` or `geom_label()`, but with one big improvement: it *repels* labels from each other and from data points, so they don't overlap. It's especially useful for labeling specific points in crowded plots.

Here's what the main parameters do:

- `data` — lets you specify a subset of your data just for labeling.
Example: `data = fish_summary %>% group_by(species) %>% slice_max(year)` labels only the most recent year per species.
- `aes(label = ...)` — tells ggplot what text to display. Usually a variable name (like `species`) or a string.
- `nudge_x`, `nudge_y` — move labels slightly along the x or y axis.
Small nudges help separate text from points or lines.

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Use Cases

Making Changes

- I want to show the maximum lengths, instead of the mean lengths. Can you give me only the section where to change that?

Sure — in your summarise step, replace `mean(length, na.rm = TRUE)` with `max(length, na.rm = TRUE)` and you can remove the standard error since it won't apply:

```
r
fish_summary <- fish %>%
  group_by(year, species) %>%
  summarise(
    max_length = max(length, na.rm = TRUE),
    .groups = "drop"
  )
```

Then in your `ggplot`, replace `mean_length` with `max_length` wherever it appears.

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Use Cases

Tedious Changes

- I want to increase all the widths and sizes by 0.5, can you change all of those.

Sure — here's your original plotting code with all **widths and sizes increased by 0.5**:

```
r
ggplot(fish_summary, aes(x = year, y = mean_length, color = species)) +
  geom_line(linewidth = 1.5) + # wds 1
  geom_point(size = 2.5) + # wds 2
  geom_errorbar(
    aes(ymin = mean_length - se_length, ymax = mean_length + se_length),
    width = 0.7 # wds 0.2
  ) +
```

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More Tools

Additional Use Cases

Use Case	Example Prompt
Commands, GitHub, command-line tools	"How do I create a GitHub repository from my existing R project? Walk me through the commands."
Theme categorization / CSVs	"Categorize these open-ended survey responses into themes and return the results as a CSV"
Learn an unfamiliar tool	"I've never used Qualtrics. Walk me through how to set up a randomized survey."
Reusable templates, functions, workflows	"Turn this analysis into a reusable R function I can use with other datasets."
Complicated or very formatted plots	"How do I combine categorical and quantitative facets in my figure in ggplot2?"
Simplify / discover code, functions, packages	"Is there a simpler way to write this R code? Are there functions or packages that could help?"
Automate messy spreadsheet cleaning	"Write an R workflow to import and clean these inconsistently formatted Excel files with messy text."
Identify patterns in figures/tables	"What patterns or notable features do you notice in this figure?"
Learn through commonly repeated asks	"Teach me regex using examples I tend to ask for in my R code."
Recreate figures from papers	"How could I recreate this figure from a paper in R?"
Transform file formats	"How do I convert this JSON file to CSV in R, and what is the difference between the formats?"
Generalize code	"Modify this code so I can use it with different variables and datasets."
Connect tools/languages	"How could I connect my Qualtrics data, R analysis, and Quarto report into one workflow?"
Test/review/refactor code	"Review this R code for errors, edge cases, and ways to simplify it."
Present results differently	"What are some different ways I could present these results?"
Find analyses/packages/tools	"What analyses and R packages are commonly used for this type of study?"

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More Tools

Additional AI Tools

Prompting

- What we've been doing

What To Try Next

- "Tools"
- Knowledge Bases/Workspaces
- Agents (OpenAI Codex, Claude Cowork, etc.)
- App Integrations (RStudio, Excel, etc.)
- APIs
- ellmer, tidyllm, rollama, chatlas

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